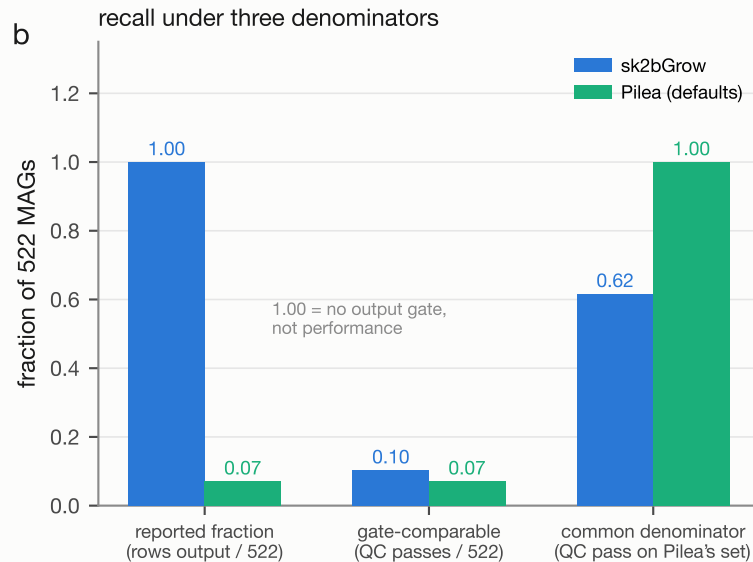
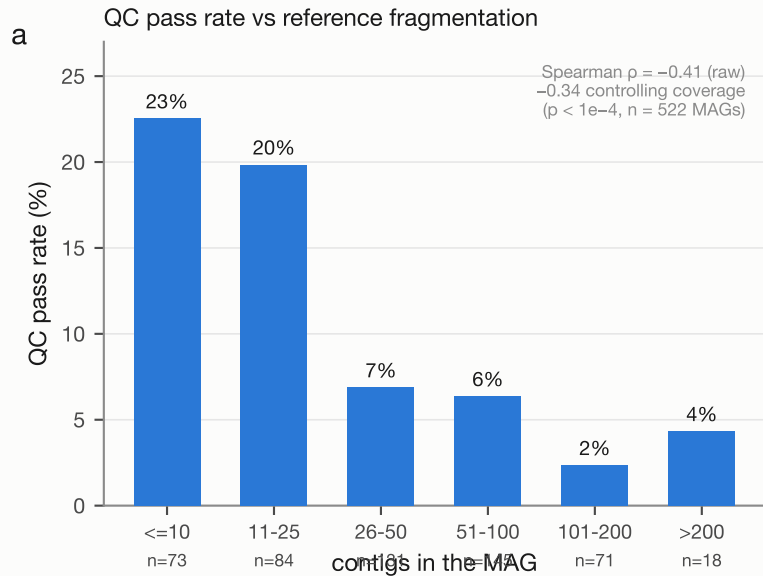




RBC metagenome (PRJNA974210): 522 MAGs $\times$ 9 samples. a, the enzyme-consistency QC passes less often as the reference fragments: mean QC pass rate falls from 22.5% (≤ 10 contigs) to 2–7% (> 25 contigs); Spearman $\rho = -0.41$ across 522 MAGs, still -0.34 after controlling for mean coverage (both $p < 1e-4$) — the QC genuinely filters fragmented references; coverage is the strongest single predictor and the two effects are partially entangled in this MAG population. b, recall under three denominators: sk2bGrow's reported fraction is structurally 1.00 (it has no output gate) — a denominator artifact, not performance; under the gate-comparable denominator (QC passes / 522) sk2bGrow reports 3.8–13.8% and Pilea (defaults) 5.0–11.1%, overlapping ranges; on the common denominator (the MAGs Pilea reports, which sk2bGrow also estimates) sk2bGrow's QC passes 43–76% of them.